

ANCHORAGE ELMENDORF AFB, AK, USA

WMO: 702720

Lat: 61.253N

Lon: 149.794W

Elev: 212

StdP: 14.58

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26401

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-9.1	-17.8	2.1	-12.4	-13.1	2.7	-7.0	25.4	30.8	20.5	28.4	5.1	60	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	74.5	59.0	71.7	58.1	68.4	56.7	61.2	70.2	59.7	67.7	58.5	65.5	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.0	72.4	61.6	56.7	69.0	60.8	55.2	65.2	60.1	27.2	70.3	26.2	67.7	25.4	65.6	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
19.8	16.6	13.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-19.4	79.2	6.4	4.2	-24.0	82.3	-27.8	84.7	-31.4	87.1	-36.1	90.2		
			-19.6	63.5	6.2	2.1	-24.1	64.9	-27.8	66.1	-31.3	67.3	-35.8	68.8		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	15.0	20.1	25.8	38.2	49.1	57.0	60.3	57.9	49.3	35.6	22.0	16.5
	DBStd	18.44	13.44	10.94	9.40	6.66	5.18	4.54	3.96	4.01	5.21	8.41	10.74	11.60
	HDD50	5503	1084	836	751	354	78	2	0	1	72	446	841	1038
	HDD65	10115	1549	1256	1216	803	493	243	155	225	471	910	1291	1503
	CDD50	879	0	0	0	1	50	212	319	244	52	1	0	0
	CDD65	16	0	0	0	0	0	4	9	3	0	0	0	0
	CDH74	109	0	0	0	0	2	30	52	25	0	0	0	0
	CDH80	9	0	0	0	0	0	2	5	2	0	0	0	0
Wind	WSAvg	5.7	5.7	5.7	6.4	6.1	6.7	6.4	5.8	5.2	5.0	5.1	5.3	5.5
Precipitation	PrecAvg	16.4	0.8	0.8	0.7	0.5	0.7	1.1	1.8	2.8	3.0	1.9	1.2	1.1
	PrecMax	27.6	2.7	2.4	2.8	1.9	1.9	3.4	4.5	9.8	7.7	4.3	2.8	2.7
	PrecMin	8.1	0.0	0.1	0.0	0.0	0.0	0.2	0.2	0.3	0.7	0.4	0.0	0.1
	PrecStd	3.5	0.6	0.5	0.5	0.5	0.4	0.7	1.0	1.8	1.7	0.9	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	43.4	47.9	61.3	72.0	77.4	79.3	77.1	66.0	54.5	46.1	45.4
		MCWB	37.0	36.7	38.9	47.4	53.2	59.7	60.9	59.7	54.0	46.2	39.8	38.6
	2%	DB	39.3	39.3	44.3	55.1	66.7	73.2	74.5	72.7	62.8	50.9	41.0	38.7
		MCWB	35.9	35.3	36.8	43.3	51.6	57.5	59.8	59.4	52.4	45.0	37.7	35.4
	5%	DB	35.7	36.5	41.0	52.2	63.2	70.0	71.6	68.9	59.9	48.3	37.5	34.9
		MCWB	33.5	33.9	34.8	41.5	49.4	56.0	59.0	58.1	51.6	43.9	35.3	33.0
	10%	DB	32.3	34.0	38.3	49.6	59.6	66.4	68.3	65.9	57.3	46.1	35.6	31.9
		MCWB	31.0	31.9	33.6	40.4	47.8	54.5	57.8	56.8	50.8	42.6	34.0	30.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.1	40.6	48.6	54.9	61.6	63.5	63.0	56.4	49.1	41.0	38.9
		MCDB	41.9	40.8	45.8	59.8	68.4	74.8	73.5	72.4	60.9	52.0	43.9	43.5
	2%	WB	35.8	36.2	37.7	44.8	52.3	59.1	61.5	60.9	54.6	46.6	38.4	35.6
		MCDB	39.2	38.9	42.0	53.0	63.6	70.5	70.7	68.5	59.0	48.8	40.9	38.4
	5%	WB	33.5	34.1	35.9	42.4	50.6	57.3	60.1	59.5	53.4	45.1	36.0	33.0
		MCDB	35.6	36.5	40.0	50.4	61.0	66.9	68.2	66.1	57.7	47.3	37.7	34.6
	10%	WB	30.9	31.7	34.1	40.8	49.0	55.6	58.9	58.2	52.2	43.2	33.8	30.5
		MCDB	32.3	33.8	37.9	48.3	58.3	64.6	66.1	63.9	55.9	45.6	35.1	31.6
Mean Daily Temperature Range	5% DB	MDBR	13.5	15.0	18.0	17.2	17.1	15.3	13.4	13.9	13.5	12.5	12.2	12.9
		MCDBR	13.6	14.6	17.8	21.0	23.2	21.7	19.2	19.7	17.1	13.8	11.8	14.1
		MCWBR	10.0	10.8	11.5	11.7	11.8	9.9	9.0	9.4	9.4	8.7	8.3	10.8
	5% WB	MCDBR	12.8	13.4	15.3	19.6	20.0	18.7	15.9	16.0	13.8	11.1	10.9	12.7
		MCWBR	9.8	10.1	10.0	11.4	10.6	9.1	8.0	8.7	8.3	7.8	8.0	10.0
Clear-Sky Solar Irradiance	taub	0.231	0.284	0.304	0.345	0.364	0.356	0.368	0.354	0.316	0.295	0.240	N/A	
	taud	2.128	2.238	2.270	2.250	2.317	2.420	2.416	2.463	2.557	2.470	2.124	N/A	
	Ebn at Noon	161	221	259	268	272	277	269	264	258	221	150	N/A	
	Edh at Noon	12	21	28	35	36	33	32	28	22	16	11	N/A	
All-Sky Solar Radiation	RadAvg	90	310	823	1363	1639	1688	1502	1189	773	358	132	47	
	RadStd	13	43	87	119	145	107	128	128	106	43	20	6	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	+4.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		N/A	+3.56	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page